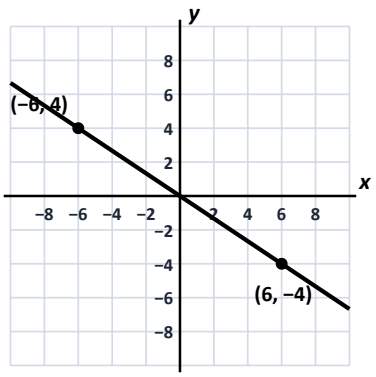
Exit Ticket

Module 2, Topic 2, Lessons 1–3: Transformations, Parallel and Perpendicular Lines

Name _____ Date _____ Period _____

1. The graph of linear function m is shown on the grid. Given f(x) = x and m(x) = a · f(x), what is the value of a?

a =

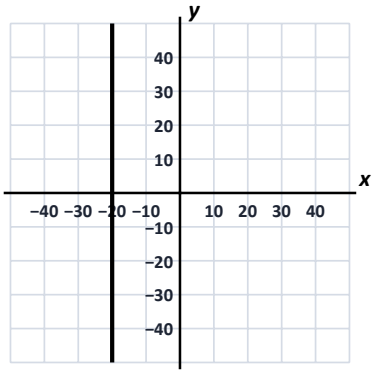


2. A tutoring center charges a fixed fee of $40, plus $15 per hour, to tutor a student for x hours. The center tutors a student a maximum of 5 hours per day. For one day of tutoring, what is the range of the function for this situation?

A. 15 ≤ y ≤ 115
B. 40 ≤ y ≤ 115
C. 0 ≤ x ≤ 5
D. 0 ≤ x ≤ 115

3. What are the equation and slope of the line shown on the grid?

F. x = −20; slope is undefined
G. y = −20; slope is zero
H. x = −20; slope is zero
J. y = −20; slope is undefined



4. Jordan graphed f(x) and g(x) = f(x − 3) on the same coordinate grid. Which statement describes how the graphs of f and g are related?

A. The graph of f(x) is translated up 3 units to produce g(x).
B. The graph of f(x) is translated down 3 units to produce g(x).
C. The graph of f(x) is translated right 3 units to produce g(x).
D. The graph of f(x) is translated left 3 units to produce g(x).

5. What is the y-intercept of the linear function created by translating f(x), represented in the table, 2 units right?

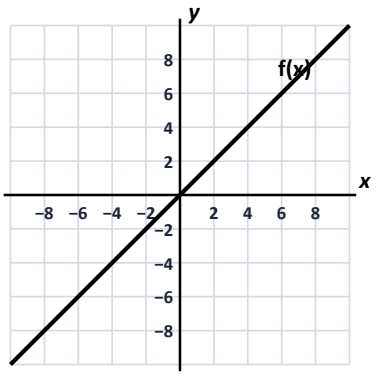
y-intercept =

x	f(x)
−2	7
−1	5
0	3
1	1
2	−1
3	−3

6. Graph g(x) based on the given transformation of f(x). Then write the equation of g(x).

f(x) = x g(x) = f(2x)

Equation of g(x):



7. What is the equation in slope-intercept form of the line that crosses the x-axis at 6 and is perpendicular to the line represented by $y = -\frac{3}{2}x + 1$?

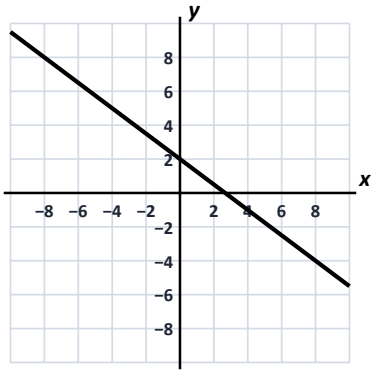
y =

8. What is the equation in slope-intercept form of the line that passes through the point (−2, 0) and is parallel to the line represented by y = 1.5x + 4?

y =

9. Write an equation in point-slope form that represents the graph.

Equation:



10. The graph of a linear function is shown on the coordinate plane. Which statements describe the graph of the linear function? Select **TWO** correct answers.

F. The graph is decreasing.

G. The slope of the line is 2.

H. The y-intercept is (0, −3).

J. The zero of the function is −3.

K. The zero of the function is 6.

